WORKED SOLUTION

STEP I 2009, Question 13

Probability and Statistics • Arrangements and Runs

STEP I 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q13

The key idea

Only the *positions* of the girls matter, not which girl or boy is which. So rather than juggling $(n+3)!$ orders, we choose the set of three places the girls occupy: every one of the $\binom{n+3}{3}$ sets is equally likely. Counting the favourable sets is then easiest if we line up the n boys first and ask which of the gaps between and around them the girls go into.

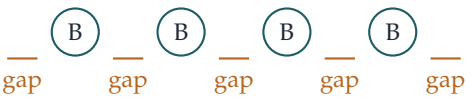
Solution

Since all $(n+3)!$ orders are equally likely, and each set of three positions for the girls arises from the same number, $3!\,n!$, of orders, the set of girls' positions is uniformly distributed over the

$$\binom{n+3}{3} = \frac{(n+3)(n+2)(n+1)}{6}$$

possible sets. Throughout, K can only be 1, 2 or 3.

The n boys in a row create $n+1$ gaps (one before the first boy, one after the last, and $n-1$ between neighbours):



Placing girls into gaps (any number into one gap) gives a line, and every line arises in exactly one way. Girls in the same gap sit next to each other; girls in different gaps are separated by at least one boy.

Part (i)

$K = 3$ means all three girls are in one block, that is, they all go into the same gap. There are $n+1$ choices of gap, so

$$\mathrm{P}(K=3) = \frac{n+1}{\binom{n+3}{3}} = \frac{6(n+1)}{(n+3)(n+2)(n+1)} = \frac{6}{(n+2)(n+3)}.$$

(Equivalently: a block of three consecutive places can start at any of positions $1, 2, \ldots, n+1$.)

Part (ii)

$K = 1$ means no two girls are adjacent, that is, the girls are in three *different* gaps. We choose 3 of the $n+1$ gaps, in $\binom{n+1}{3}$ ways, and put one girl in each. Hence

$$\mathrm{P}(K=1) = \frac{\binom{n+1}{3}}{\binom{n+3}{3}} = \frac{(n+1)n(n-1)}{(n+3)(n+2)(n+1)} = \frac{n(n-1)}{(n+2)(n+3)},$$

as required. (For $n = 0$ or 1 there are too few gaps, and the formula correctly gives 0.)

? **Ask yourself** Why is the count $\binom{n+1}{3}$ and not $\binom{n+1}{3} \times 3!$?

We are counting sets of *positions*, not orders of named children. Choosing which three gaps are used fixes the girls' positions completely; which girl sits where has already been accounted for in the claim that every set of positions is equally likely.

Part (iii)

First we need $\mathrm{P}(K=2)$. Since $K \in \{1,2,3\}$,

$$\begin{aligned}\mathrm{P}(K=2) &= 1 - \frac{6}{(n+2)(n+3)} - \frac{n(n-1)}{(n+2)(n+3)} \\ &= \frac{(n^2+5n+6) - 6 - (n^2-n)}{(n+2)(n+3)} = \frac{6n}{(n+2)(n+3)}.\end{aligned}$$

As a check, count directly: $K = 2$ means a pair of girls in one gap and the third girl in a different gap. Choosing the gap for the pair and then a different gap for the single gives $(n+1)n$ sets, and $(n+1)n/\binom{n+3}{3} = 6n/((n+2)(n+3))$, as before.

Now

$$\begin{aligned}\mathrm{E}(K) &= 1 \cdot \frac{n(n-1)}{(n+2)(n+3)} + 2 \cdot \frac{6n}{(n+2)(n+3)} + 3 \cdot \frac{6}{(n+2)(n+3)} \\ &= \frac{n^2+11n+18}{(n+2)(n+3)} = \frac{(n+2)(n+9)}{(n+2)(n+3)} \\ &= \frac{n+9}{n+3}.\end{aligned}$$

Checks

- $n = 0$: there are only girls, so $K = 3$ certainly, and $\frac{n+9}{n+3} = 3$.
- $n = 1$: the lines are GGGB, GGBG, GBGG, BGGG, equally likely, so $K = 3$ or 2 with probability $\frac{1}{2}$ each and $\mathrm{E}(K) = \frac{5}{2} = \frac{10}{4}$.
- As $n \to \infty$, $\mathrm{E}(K) \to 1$: with many boys the girls are almost never neighbours.
- We enumerated all position sets for $n = 0, \ldots, 9$ with exact fractions (for example $n = 3$ gives $\mathrm{P}(K = 1, 2, 3) = \frac{1}{5}, \frac{3}{5}, \frac{1}{5}$ and $\mathrm{E}(K) = 2$), and all $(n+3)!$ orders of labelled children for $n \leq 4$; everything matched.

Common slips

- Mixing counts of ordered arrangements with counts of position sets in the same fraction. Pick one sample space and stick to it.
- Forgetting the two end gaps, and so using $n - 1$ gaps instead of $n + 1$.

Beyond STEP

The "boys make gaps, girls go into gaps" trick is the stars-and-bars idea, and the general question of the longest run in a random arrangement is a classical one: for long random sequences the longest run grows like a logarithm of the length. An undergraduate probability course meets these through indicator variables and generating functions. Grimmett and Welsh, *Probability: An Introduction*, is a good place to see such counting arguments developed.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.